

Ethan Tsao

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EDUCATION

University of California, Santa Barbara, UC3M Abroad

B.S. in Statistics & Data Science, Technology Management Program

Graduation Date: Dec. 2025

Cumulative GPA: 3.40/4.0

Coursework: Regression Analysis, Statistical Machine Learning, Stochastic Process, Databases Design, Experimental Design

SKILLS

Languages & Core Analytics: Python (Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn), SQL (PostgreSQL, MySQL, SQL Server), R, Excel (Pivot Tables, Power Query)

Visualization & BI Tools: Power BI, Tableau, Looker, Plotly, ECharts

Big Data & Cloud: Azure, AWS, GCP, Git, Docker, JavaScript, React, FastAPI, REST APIs

TECHNICAL PROJECTS

JobFlow and Market Intelligence Dashboard

Jan. 2026

- Built a full-stack job intelligence platform (React, FastAPI, PostgreSQL) processing 80+ applications through a built-in Chrome extension; built Python ETL pipeline extracting 90+ technical skills and standardizing salary data with 81% parsing accuracy.
- Designed Tableau market analytics dashboard identifying compensation patterns: mid-sized firms (500-1K employees) pay 31% premium, hybrid work commands 17% salary advantage, and Snowflake/Airflow expertise correlates with \$162K+ salaries.
- Developed 8 SQL queries analyzing role distribution, skills demand, and industry concentration, revealing Python (75%) and SQL (70%) as universal requirements while Software/Tech sector comprises 71% of data role opportunities.
- Transformed unstructured job descriptions into actionable market insights showing role-specific skill hierarchies: Data Science roles require ML (100%), Data Engineering emphasizes ETL (65%), and Data Analyst positions prioritize BI tools (71%).

Video Game Database

Nov. 2025

- Architected a normalized relational database with 13+ tables using MySQL, implementing referential integrity constraints and validation rules to maintain data quality and enable efficient inventory management queries.
- Developed 15+ complex SQL queries utilizing multi-table JOINS, subqueries, aggregations, and window functions to extract business insights and support real-time reporting requirements for inventory analytics.

Bank Churn Prediction

Mar. 2025

- Developed predictive churn models (XGBoost, Random Forest, Bagging, K-NN, Logistic Regression) achieving 86.1% accuracy through stratified cross-validation and SMOTE balancing to address class imbalance.
- Conducted exploratory data analysis with statistical visualizations (density plots, correlation heatmaps, stacked bar charts) to identify age-based churn patterns and feature relationships across 10,000 customer records.
- Enhanced model performance through systematic preprocessing pipeline including min-max scaling, one-hot encoding, and IQR-based outlier removal, resulting in 0.852 ROC-AUC score with XGBoost classifier.

WORK EXPERIENCE

Aquila Clouds

Pleasanton, CA

Software Engineer Intern

Jun. 2025 - Sep. 2025

- Refactored 20+ React components to improve scalability and reduced interface load times by 9% across FinOps dashboards. Proactively implemented robust error boundaries and data validation, increasing UI reliability and debuggability.
- Led full-stack integration of REST APIs and EChart visualizations for product feedback and month-over-month cost insight.
- Analyzed cloud cost data across 50+ customer accounts using SQL, identifying spending patterns and optimization opportunities.

Mosher Alumni Hall

Santa Barbara, CA

Alumni Affairs Intern

Oct. 2024 - Jun. 2025

- Coordinated logistics for 60+ university events, managing vendor coordination, AV setup, and attendee experience.
- Maintained event tracking system in Excel documenting attendance patterns, budget expenditures, and resource allocation.

ORGANIZATIONS

Wedge Health (YC S25)

Remote

Technical Consultant

Aug. 2025 - Oct. 2025

- Conducted targeted outreach to healthcare organizations and provided consultative support for a health AI platform, educating potential stakeholders on AI-driven patient engagement solutions and identifying integration opportunities.

Gaucha Sports Analytics

Santa Barbara, CA

Project Member

Mar. 2025 - Sept. 2025

- Collaborated with UCSB Baseball to develop lineup optimization logic using WAR, ETL pipelines, and OBP-based modeling.

Delta Sigma Pi - Rho Sigma Chapter

Santa Barbara, CA

VP of Professional Activities

Feb. 2024 - Apr. 2024

- Organized keynote presentation with NASA JPL speaker for 50+ attendees, managing speaker coordination and logistics.